



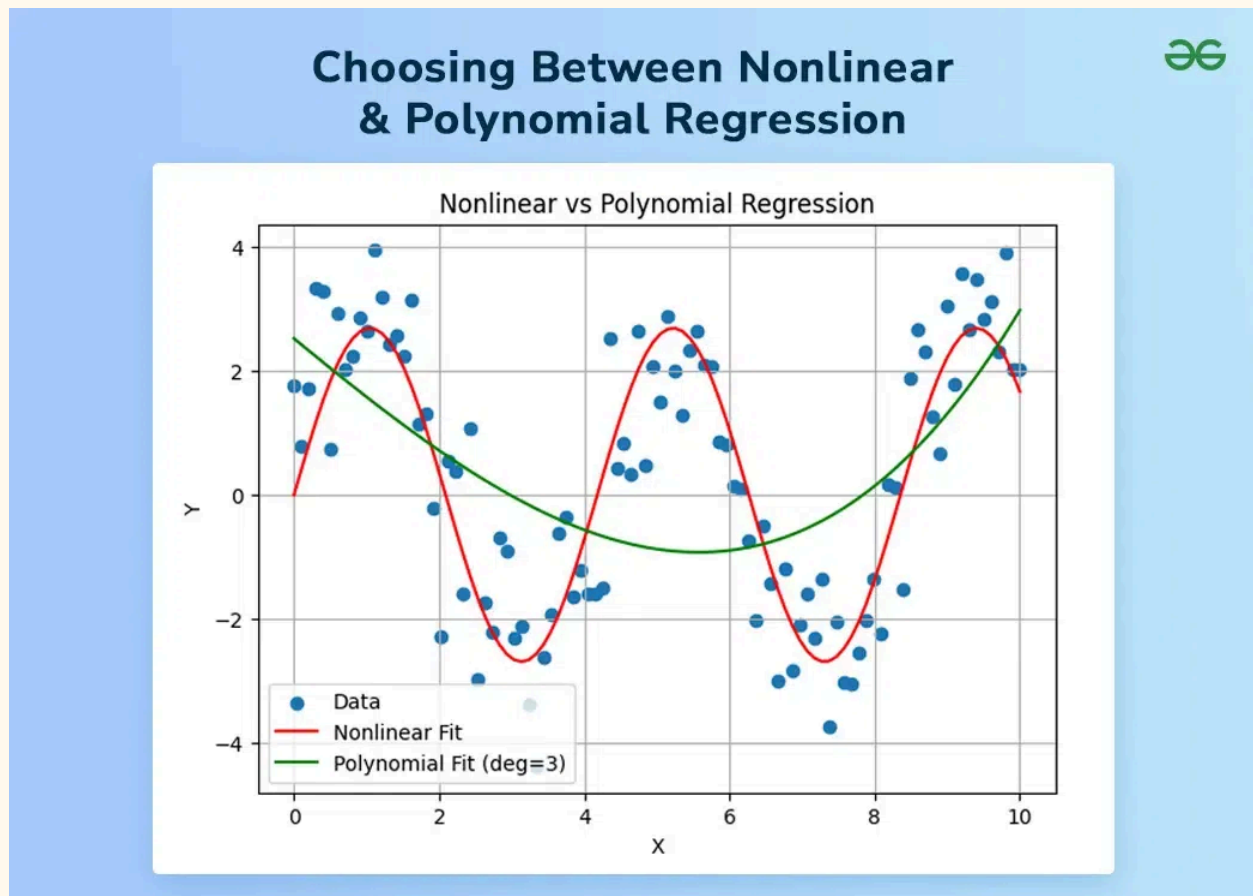
IEEE

AI Team: Task 6

Thesis

Polynomial Regression

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INTRODUCTION

Polynomial Regression is a type of regression models where instead of adjusting the weights to linear data, you adjust the weights to features of higher order.

Why? You might ask

- To capture more of the nature of the data rather than just fitting a straight line in the data, which would reduce the bias of the model.

Overfitting

As we make the models with higher degrees, we risk overfitting the model to the data, meaning: the model merely memorizes the data instead of actually learning, which makes it performs really well on data it seen but not so good on unseen data.

We fix this issue using L1 and L2 regularization.

Ridge Regularization (L2)

Ridge regression works by adding a penalty to the loss function that grows with the size of the model's coefficients. Why? Because when you use polynomial features, the model can end up assigning huge weights to certain terms just to chase noise in the training data — that's overfitting. Ridge discourages this by punishing large coefficients, effectively telling the model "keep your weights small unless you really need them to be big." It shrinks all coefficients toward zero, but rarely all the way to zero, which means every feature stays in the model, just with a dampened influence.

Lasso Regularization (L1)

Lasso does something similar to Ridge, but with a twist: instead of just shrinking coefficients, it can push some of them all the way to exactly zero. Why does that matter? Because with polynomial features, you often end up with a lot of redundant or meaningless terms — like the interaction between cylinder count and country code, which doesn't represent anything physically meaningful. Lasso automatically detects which of these terms aren't pulling their weight and eliminates them entirely, effectively doing feature selection for you as part of training.

Few metrics to check how a model is performing

Mean Squared Error (MSE)

MSE measures the average of the squared differences between predicted and actual values. Why square the errors? Because squaring punishes large mistakes disproportionately more than small ones — a prediction that's off by 10 contributes 100x more to the error than one off by 1. This makes MSE sensitive to outliers, which is useful if large errors are especially costly, but risky if your data has a few extreme values that could distort your sense of overall performance.

Mean Absolute Error (MAE)

MAE takes the average of the absolute differences between predicted and actual values, without squaring them. Why does this matter? Because unlike MSE, it treats all errors proportionally — being off by 10 is exactly 10x worse than being off by 1, no more, no less. This makes MAE more robust to outliers and easier to interpret directly in the same units as your target variable (in this case, miles per gallon).

Root Mean Squared Error (RMSE)

RMSE is just the square root of MSE. Why bother taking the square root? Because MSE is in squared units (e.g. mpg^2), which isn't intuitive to interpret. Taking the square root brings it back into the original units (mpg), while still preserving MSE's sensitivity to large errors. This makes RMSE a good middle ground — interpretable like MAE, but still penalizing big mistakes more heavily.

Mean Absolute Percentage Error (MAPE)

MAPE expresses error as a percentage of the actual value, rather than in raw units. Why is that useful? Because it lets you compare model performance across different scales — a 2 mpg error means very different things for a 10 mpg car versus a 40 mpg car, and MAPE captures that relative difference directly. The catch is that it becomes unstable or misleading when actual values are close to zero, since dividing by a tiny number blows up the percentage.

R² (Coefficient of Determination)

R² tells you what proportion of the variance in your target variable is explained by the model, on a scale that tops out at 1.0 for a perfect fit. Why is this different from the error metrics above? Because R² isn't measured in the same units as your target — it's a relative score, telling you how much better your model is than simply predicting the average every time. An R² of 0.91, for example, means your model explains 91% of the variation in mpg. The limitation is that R² can be misleadingly inflated by adding more features (even useless ones), which is exactly why comparing R² alongside error metrics like RMSE matters — it keeps the model honest.

THE CONCLUSIONS ARE IN THE NOTEBOOK FILES IN CODE